Explanation Of Revisions

Title: Fight Fire with Fire: Adaptive Black-Box Defences for Misinformation Detection

ACL ARR 2026 May Submission 3459 -> August Resubmission

We would like to thank the reviewers and the area chair for the effort put into reviewing our contribution. While the paper in its current form is currently border-line recommended for acceptance into Findings, we are going to take the opportunity for further improvement.

A. Responses to Area Chair

Below you will find the key revisions suggested by the AC with our explanation on how we address them. At the end of this document we attach a version of our contribution including highlighted changes.

1. Estimating true label for reinforcement learning reward

As reviewers M6fk and eKBm point out, our initial MACABEU-online method was limited by the fact that the reward function depended on the true gold label, which might not be known in practice. Therefore, we compare the offline approach to:

- MACABEU-oracle, using the true gold label as previously, if it is available for adaptation, for example coming from human feedback.
- MACABEU-estimated, where the true label is estimated using 7-fold voting (one of the defense mechanisms). This allows us to proceed even if the gold label is unknown.

In section 3.3 we describe the two new MACABEU variants. We have added a new Table 9 in Appendix E, comparing their performance on BiLSTM and BERT. The estimated variant preserves most of the online adaptation gain: mean BODEGA is only ~20% higher than oracle on the BiLSTM+BERT sub-grid, and still below every static defence in Table 1. On character-level attackers (BERTattack, XARELLO) it comes within 5% of oracle; on word-level ones (PWWS, Genetic) the gap is wider (+60% and +27%), a known limitation of consensus-based pseudo-labels that we discuss in Limitations (x). We believe this is a valuable improvement, making the proposed approach directly applicable.

2. Type of RL paradigm

As reviewer eKBm pointed out, we claimed to have used Q-learning, but the actual method employed (contextual bandit) does not fall under this paradigm, as it disregards future rewards.

This was indeed a mistake, and it has been corrected. The relevant section (3.3) was amended accordingly.

3. Justification of query capping

Our experiments limit the number of queries to 500, making it difficult to attack examples with longer text. This aspect was commented on by reviewer LQQa.

While we acknowledge it makes attackers' role more difficult, we need to limit the query budget to make the procedure realistic. No public platform would allow its users to make thousands of attempts to publish a post. This limitation has also been introduced in previous work, with even lower quotas than ours. We now provide a justification and reference to relevant work in section 3.1.

4. MACABEU feature importance

MACABEU uses a variety of features to decide which defence is appropriate for a particular example. As reviewer M6fk notes, we do not perform experiments to find out which of those are indeed relevant for the task.

To alleviate this limitation, we analyse the contribution of each of them by checking how they influence the choices made by MACABEU. The whole experiment and its results are described in appendix F, with a shortened conclusion in section 5.3. Four features dominate the policy's decisions across all trained variants: text length, uppercase ratio, word count, and OOV ratio; the remaining six features have flip rates below 0.11 and barely influence action selection.

5. Framing problem

Indeed, the fragments of the introduction that made it look like there is a "lack of black box defense methods" and we address it by designing a new technique, might appear misleading. We have clarified it by rewording several fragments, especially the contributions at the end of the introduction section in order to make it clear that we provide:

- A systematic analysis of how the already existing black-box defences fare in the misinformation detection use cases,
- A meta-defender that combines these defences to find the best approach for each adversarial example.

In the related work section we also review a plethora of previous publications on black-box defences.

B. Other Changes

Apart from the changes above, we have improved the manuscript in the following ways:

6. Availability of source code

As reviewer eKBm pointed out, the anonymised source code link made it impossible for the reviewers to access it. In order to make the code available while keeping the anonymity requirements, we attach the code to the submission.

7. Clean accuracy scores

Our measurements of clean accuracy (when no attack is performed) in Table 2 and Figure 3 included static defences and MACABEU-off. Reviewer M6fk pointed out that the results should also cover MACABEU online.

This was improved by adding the two online variants (MACABEU-oracle and MACABEU-estimated) to the clean accuracy results. Both are competitive: MACABEU-oracle costs only −0.6% clean accuracy relative to baseline while cutting mean BODEGA to 0.097, and MACABEU-estimated costs −4.0% while still dominating every static ensemble defence on the accuracy–robustness plane (Figure 3).

8. Other changes

We have also corrected further small mistakes and improved editing to fit in the page limit despite expanding the content to address the reviewers' concerns.

9. Lack of analysis of cross-architecture differences, larger victim models or attackers adapting to online MACABEU

We are sorry, but the limitations of both the length of this paper (which already includes numerous appendices providing additional information), as well as the extensive computational cost of training attackers and defenders, mean that these additional experiments will have to remain out of scope. Our study provides the first comparison of black-box defences in literature: systematically across tasks, victim architectures, and attackers, as well as a learned per-example defence selector. We hope to explore the further extensions to this seminal study in future work.

D. Change-Highlighted Version of the Manuscript

(see below)